

Rohan Jain

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SUMMARY

I build LLM systems that hold up in production. At Bright Machines I work on an on-prem AI platform for factory software: LangGraph agents calling 132 MCP tools through a LiteLLM gateway, and the FastAPI and React/TypeScript services around them on Kubernetes. I built the evaluation harness behind the agent's tool-selection plan (recall 0.47 to 0.97) and fixed a 74-minute silent outage that no health check detected. MS Data Science, CU Boulder, GPA 3.9, May 2026.

SKILLS

LLM & Agents: Python, LangGraph (ReAct, multi-agent), LangChain, Model Context Protocol (built and hardened MCP servers), LLM tool-calling agents, RAG (BM25, embeddings, hybrid), Pydantic v2 structured outputs, LiteLLM multi-provider gateway, OpenAI / Anthropic / Gemini APIs, LLM evaluation harnesses, prompt-injection testing, guardrails

Backend & Data: FastAPI, PostgreSQL + Alembic, SQL, SQLite / sqlite-vec, Databricks (SQL, Unity Catalog, Asset Bundles), Temporal, Docker, Kubernetes / Kustomize, ArgoCD, GitLab CI

Frontend, Cloud & Observability: TypeScript, React, Vite, Tailwind; Azure (AKS, Entra ID), AWS (EC2, Lambda, S3, API Gateway, RDS); OIDC auth (oauth2-proxy, Keycloak); Langfuse tracing, Grafana, Prometheus, MLflow

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 - Present | San Francisco, CA

- Built a 115-question evaluation harness for an on-prem LLM agent whose 132 MCP tool schemas overflowed its 32K-token window (LangGraph ReAct over LiteLLM), and a tool-selection plan measured to lift recall from 0.47 to 0.97.
- Designed and shipped a read-only MCP server from scratch (37 files, +3,163 lines, 12 tools) that lets the agent query a 32,000-workflow production Temporal namespace, deployed on Kubernetes with Kustomize.
- Hardened four MCP servers: closed a Critical path-traversal vulnerability (CWE-22), added least-privilege RBAC and a fail-closed Grafana allow-list (51 tools to 16), and proved by prompt injection the agent cannot write to the database.
- Shipped a self-healing watchdog, verified live on a real gateway restart, after root-causing a 74-minute silent chat outage no health probe detected; grew the orchestrator's test suite from 0 to 135 tests in CI.
- Built an on-prem grounded document Q&A platform end to end (FastAPI, React/TypeScript, sqlite-vec retrieval, OpenAI-compatible API, fail-closed OIDC seam) with 352 automated tests, cutting fabricated citations from 31% to 0%.
- Grounded the factory's LLM-written daily production report in the dashboard's API (now in production): six sections no longer come from SQL tools the model chose to call, correcting a yield figure filtered on an outdated station.
- Caught an LLM merge step silently dropping up to 12% of grounded facts while all 30 tests passed, and fixed it by re-grounding the merged output against the extracted facts, recovering every dropped fact on two real runs (6/6 and 1/1).

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 - May 2026 | Boulder, CO

- Replaced a legacy SVM bias-correction model with a tuned Random Forest for NWP post-processing (feature engineering, cross-validation), cutting temperature MAE 47% and humidity MAE 54%; promoted to production.
- Ran controlled loss-function ablations (CRPS vs hybrid CRPS+MSE vs Huberized CRPS); the hybrid trained 17% faster with a consistent CRPS gain, and the results informed the production loss choice.
- Built an automated inference quality gate that rejects any model failing to beat the pre-ML baseline, cutting failed production runs 90% and saving about 20 engineer-hours a month.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 - May 2026 | Boulder, CO

- Migrated lab infrastructure to AWS EC2 behind Keycloak OIDC with Grafana/Loki/Prometheus observability; reverse-engineered proprietary Thermo Isodat binaries into deterministic R readers with golden-output regression tests.

EDUCATION

University of Colorado, Boulder

Aug 2024 - May 2026 | Boulder, CO

Master's in Data Science (GPA: 3.9/4.0)

Mumbai University, Rajiv Gandhi Institute of Technology

Jul 2020 - May 2024 | Mumbai, India

B.E. in Computer Engineering (GPA: 3.7/4.0)

PROJECTS

ResearchAgent (Multi-Agent Research Assistant) | LangChain, FastAPI, SSE

Jun 2026

- Four LangChain agents (researcher, summarizer, critic, reporter) stream progress to a React UI over Server-Sent Events, with structured JSON logs of every LLM and tool call; live demo at about \$0.01 to \$0.05 per query.

Flan-T5 QLoRA Fine-Tune | PyTorch, Hugging Face PEFT, LoRA

Jun 2026

- Fine-tuned flan-t5-base with QLoRA (8-bit, rank 16) on a 360-example synthetic QA set, lifting ROUGE-L from 0.1637 to 0.2058 (+25.7%) in 2.2 minutes on a free T4; adapter published to Hugging Face Hub.

ClaimPilot AI (Guardrailed Tool-Calling Agent) | Python, Pydantic v2, OpenAI API

Jan 2026

- Deterministic Pydantic v2 checks find healthcare-claim errors and OpenAI function calling proposes fixes only for those; strict schemas with retry and fallback guarantee a valid report even if the model fails.